IC104 Introductory Quantitative Techniques for Business and Finance

Lecture 4. Topics in Financial Economics

Dr Nikolaos Antypas

ICMA Centre
Henley Business School
University of Reading

Lecture Plan

- Textbook Reading (Sydsaeter & Hammond Chapters 11.1-11.7)
  - Rate of Return
  - Geometric Series
  - Time Value of Money

Percentages and Rate of Return

- A percentage is a number expressed as a fraction of 100. The percent value is calculated by multiplying the numeric value of the ratio by 100. We denote a percentage by x%.
  - Thus, all these are equivalent: $20\% = \frac{20}{100} = 0.2$
  - Percentages are used to refer to proportional changes of a variable over time.
- In economics and finance, we often are interested in analysing variables that increase or decrease over time. How to calculate the percentage changes?

$$Change = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}}$$

Example

- Suppose that the stock price of Microsoft was \$30 in January 2013, while it was \$25 in January 2012. What is the percentage change of Microsoft's stock price from 2012 to 2013?

$$\frac{30-25}{25} = 0.2 = 20\%$$

- Suppose that the stock return of Amazon is 45% over year 2012. The stock price at the beginning of 2012 was \$170. What is the Amazon stock price at the end of 2012?

$$\frac{S-170}{170} = 45\% \Rightarrow S = (170 * 0.45) + 170 = \$246.50$$

Geometric Series

- A geometric series is a sequence of numbers where each new number is generated by multiplying the previous number by a constant term.
  - The sequence $1, 2, 4, 8, 16, ...$
  - This year, a firm has annual revenue of \$100 million that it expects to increase by 16% per year throughout the next decade. What are the firm's annual revenues in each of the next 12 years?:
  - 100, $100 * (1 + 16\%)$, $100 * (1 + 16\%)^2$,...,$100 * (1 + 16\%)^{11}$,
- The constant term is called the "Common Ratio", "Geometric Ratio", or "quotient".
- The general form of a geometric series is:

$$a, ak, ak^2, ..., ak^n, ak^{n+1}, ...\text{where k is the common ratio}$$

- The sum of the first n terms of a geometric series is:

$$s_n = a + ak + ak^2 + \cdots + ak^{n-2} + ak^{n-1}$$

Summation Formula for A Finite Geometric Series

Theorem

The sum of the first n terms of a geometric series: $s_n = a\dfrac{k^n - 1}{k - 1}$ $(k \neq 1)$

Proof.

1. Multiply k on both sides: $s_n = a + ak + ak^2 + \cdots + ak^{n-2} + ak^{n-1}$ (A)
2. $ks_n = ak + ak^2 + ak^3 + \cdots + ak^{n-1} + ak^n$ (B)
3. $B - A \Rightarrow ks_n - s_n = ak^n - a \Rightarrow s_n = a\dfrac{k^n - 1}{k - 1}$ □

- Example: $100 + 100 \cdot 1.16 + \cdots + 100 \cdot (1.16)^9 = 100\dfrac{(1.16)^{10} - 1}{1.16 - 1} = 2132$

Summation Formula for An Infinite Geometric Series

Theorem

The sum of a geometric series with infinite terms: $s = \frac{a}{1-k}$ $(|k| < 1)$

Proof.

1. Multiply k on both sides: $s = a + ak + ak^2 + \cdots + ak^{n-1} + ak^n + \cdots$ (A)
2. $ks = ak + ak^2 + ak^3 + \cdots + ak^n + ak^{n+1} + \cdots$ (B)
3. $B - A \Rightarrow ks - s = -a \Rightarrow s = \frac{a}{1-k}$
4. The other way, $s = \lim_{n\to\infty} a\frac{k^n - 1}{k-1} = \frac{a}{1-k}$ $(|k| < 1)$ □

- Example: $1 + 1 \cdot 0.25 + 1 \cdot (0.25)^2 + 1 \cdot (0.25)^3 + \cdots = \frac{1}{1-0.25} = \frac{4}{3}$

The Time Value of Money

- Easy choice: £100 or £110 right now.
- Harder to decide: £100 today or £110 in 1 year.
- Fact: A dollar today is usually worth more than a dollar tomorrow, unless interest rates are negative.
  - Future Value: How much is one dollar today worth in the future?
  - Present Value: How much is one dollar in the future worth today?
- Why is that?
  - Interest! We can take the dollar and invest it and earn interest.
  - In economists' words, the price of holding cash is interest.
  - A dollar in 1913 is different from a dollar in 2013.
- The Time value of money is **the difference in value between money today and money in the future**

The Time Value of Money

- The Time Value of Money is one of the central concepts in Finance.
- Only some finance professionals and researchers use derivatives, but nearly all have to calculate the time value of money.
- What is the fair value of a cashflow at some future point? We need to know in order to value companies, as the fair value of a share is the expected Present Value of its future dividends. We also need to be able to calculate this in order to value Bonds (e.g. Government Bonds)

The Time Value of Money

- What is a given sum of money worth in the future, given an interest rate? We need this to be able to calculate no-arbitrage values of Foreign Exchange rates (FX rates), given the interest rates in two countries.
- If a project or a bond costs us a known amount of money today and has a predictable set of future cashflows, what is the effective interest rate (called the Internal Rate of Return or IRR) for the project / bond? This allows us to see whether a project is worth the cost of the capital used to fund it, or to compare the returns on two different bonds.

Interest Rate

- If you put \$100 in the bank at 6% interest, how much would you have in one year if the interest were paid annually?

$$\$100 * (1 + 6\%) = \$106$$

- If you put \$100 in the bank at 6% interest, how much would you have in three years if the interest were paid annually?

$$\$100 * (1 + 6\%)^3$$

- If you put \$100 in the bank at 6% interest, how much would you have in one year if the interest is paid semi-annually?

$$\$100 * (1 + \frac{6\%}{2})^2$$

- If you put \$100 in the bank at 6% interest, how much would you have in one year if the interest is paid quarterly?

$$\$100 * (1 + \frac{6\%}{4})^4$$

Effective rate of interest

- Effective rate of interest: When interest is paid n times during the year at rate $\frac{r}{n}$ per period, then the effective annual rate R is defined as:

$$R = (1 + \frac{r}{n})^n - 1$$

- When investing in a savings account, which of the following offers are better: 5.9% with interest paid quarterly; or 6% with interest paid twice a year?

$$\text{Effective Rate of Option 1} = (1 + \frac{0.059}{4})^4 - 1 \approx 0.0603$$

$$\text{Effective Rate of Option 2} = (1 + \frac{0.06}{2})^2 - 1 \approx 0.0609$$

Comparing Different Interest Periods

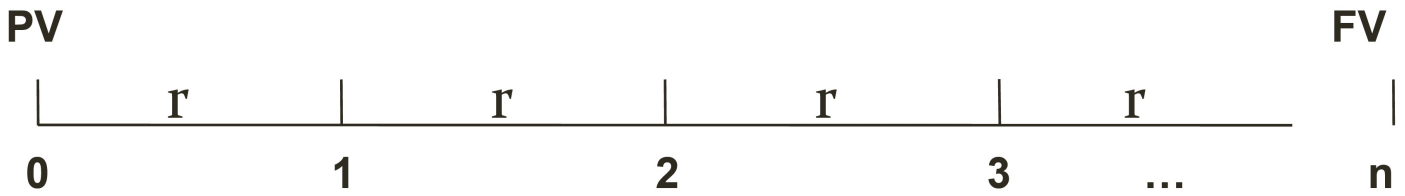
- Given annual interest rate of 10%

Compounding Interval	Number of Times (n)	Effective Interest Rate
Year	1	10.00000%
Quarter	4	10.38129%
Month	12	10.47131%
Week	52	10.50648%
Day	365	10.51558%
Hour	8,760	10.51703%
Minute	525,600	10.51709%

Comparing Different Interest Periods

- What happens if n goes to infinity?
  - The Effective interest rate becomes $e^{rt} - 1$
  - Notice that the Exponential function has appeared.
  - One of the reasons that the Exponential function is used so much in the mathematics finance (especially Derivatives valuation and Bond pricing) is that e^{rt} represents the continually-compounded interest rate. This is why it is so important to become familiar with the mathematics of the Exponential function.
  - For one year, at 10% continuous compounding: $e^{0.1 \cdot 1} = 0.105170918$

Time Value of Money



- Consider the time line above: we start with a sum of money which is worth PV at time $t = 0$, and which receives interest at rate r over n periods. At the end, at $t = n$, it is worth FV.
- PV is the Present Value.
- FV is the Future Value.
- The number of time periods between the Present Value and the Future Value is represented by "n" or "N".
- The rate of interest is called "r".
- Given any three of PV, FV, r and n, we can calculate the fourth.

FV, PV, r and n

- Future Value: The amount an investment is worth after one or more periods at some given interest rate:

$$FV = PV(1 + r)^n$$

- The present value is the current value of future cash flows discounted at the appropriate discount rate. Solving for PV from the FV formula:

$$PV = \frac{FV}{(1 + r)^n}$$

- The effective interest rate over one time period, given a Present Value of PV at time $t = 0$ and a Future Value of FV at time $t = n$:

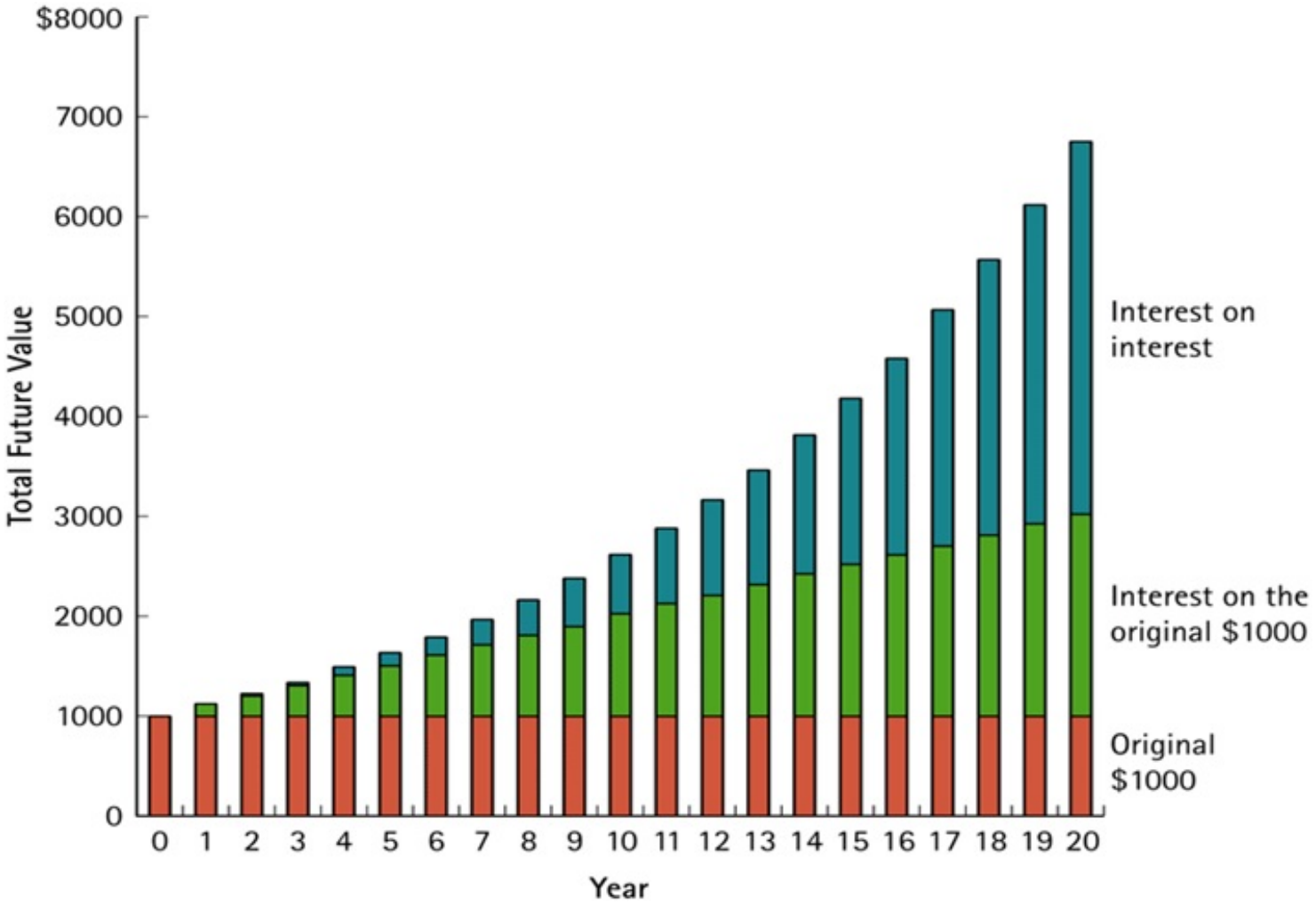
$$r = [(\frac{FV}{PV})^{\frac{1}{n}} - 1]$$

FV, PV, r and n

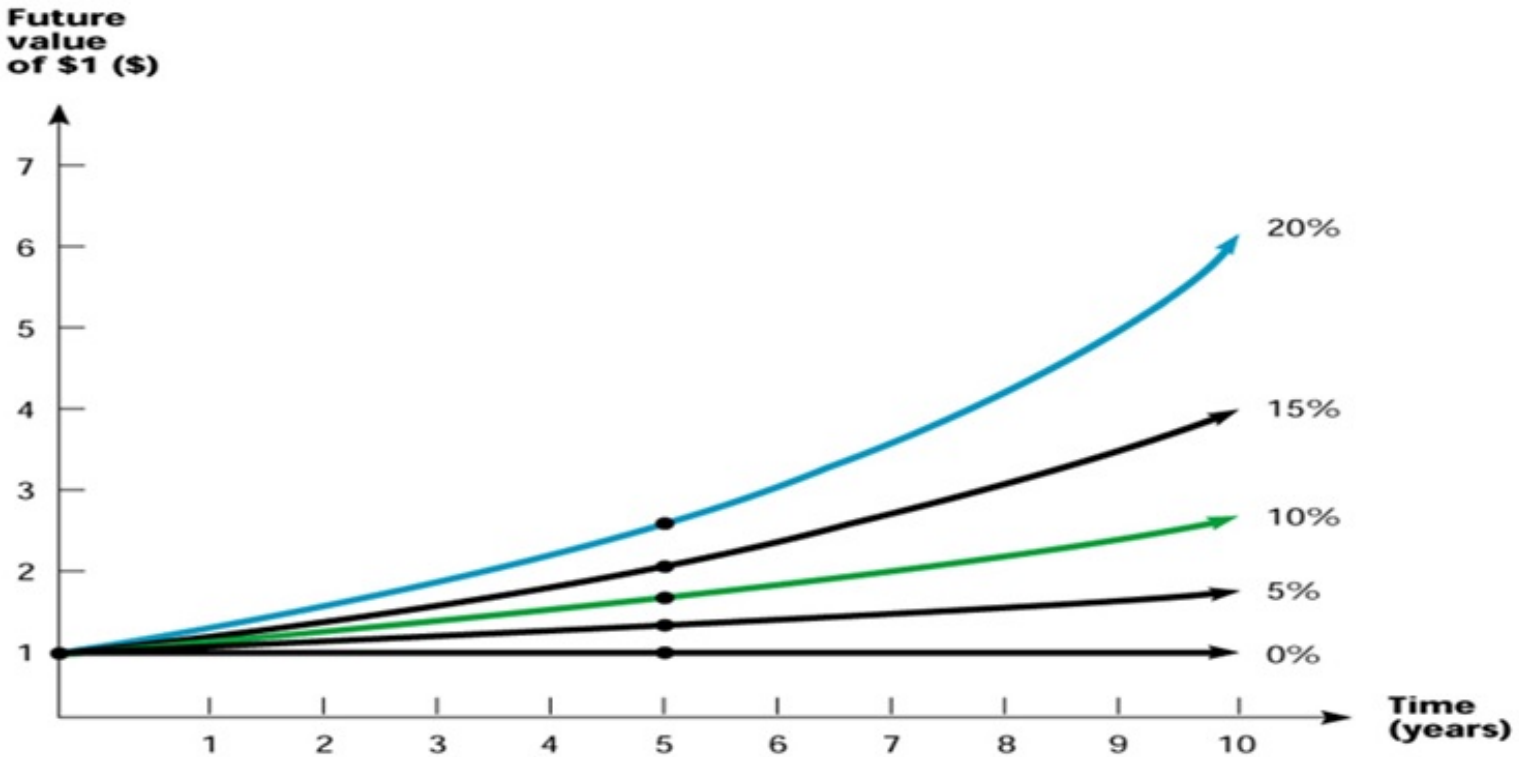
- n, the number of periods required for an initial investment of PV to grow to FV given an interest rate of n:

$$n = [\frac{\ln(\frac{FV}{PV})}{\ln(1+r)}]$$

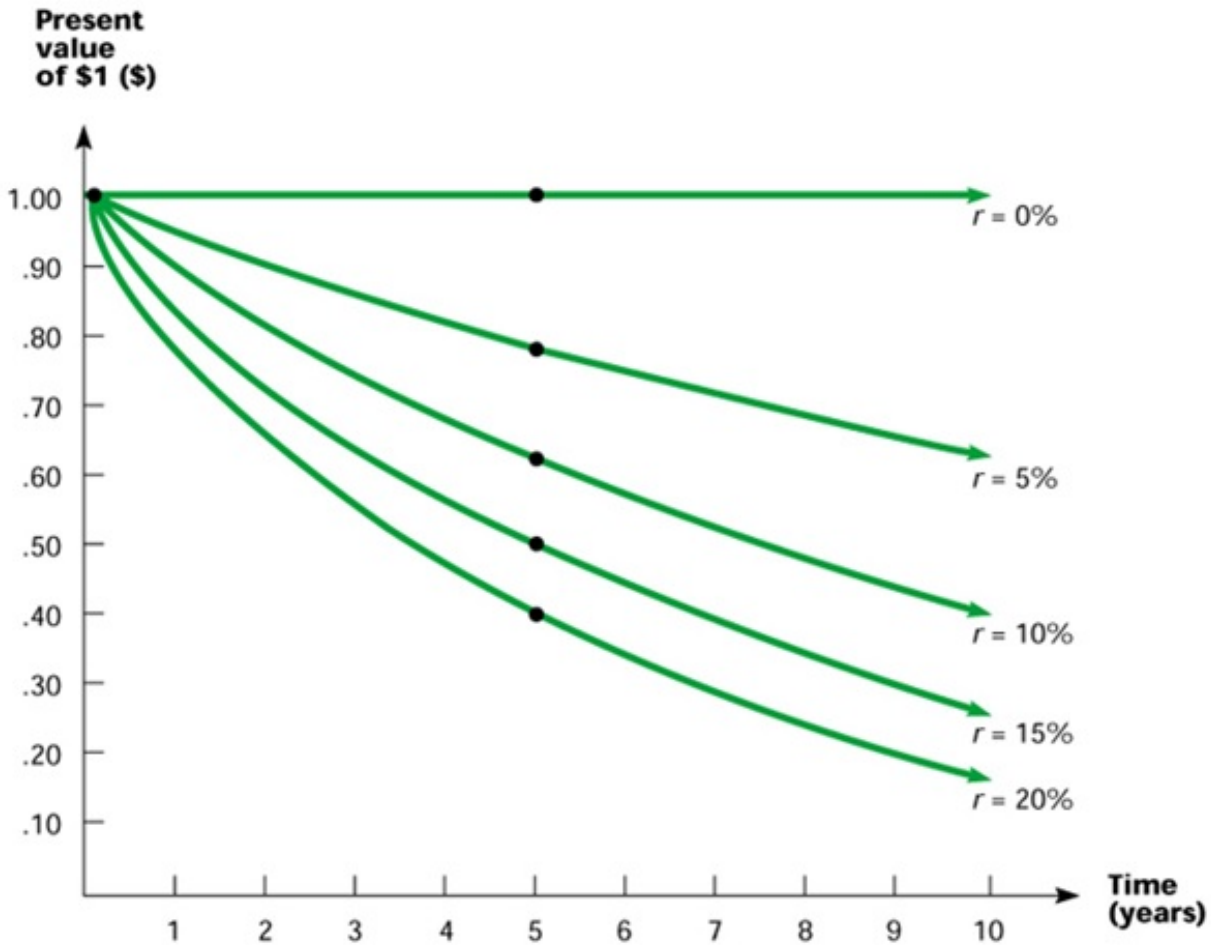
The Composition of Interest over Time



Future Value vs. Interest Rate



Present Value vs. Interest Rate



Example

- Iago the Innocent has 100 dollars to invest for one year. Hamlet offers him a monthly compound rate of 5%. How much should Hamlet return at the end of the year?

  Hamlet's Offer: $FV = PV(1+r)^n = 100 \cdot [(1+5\%)^{12}] = 179.5856$

- Megacafe plans to launch an advertising campaign to promote their new drink, the Velvet Hammer (six shots of espresso and one shot of milk.) They plan on rolling out the product two years from now. How much will they need to invest today in order to have 2 million for advertising, assuming that the annual interest rate is 10 percent?

$$PV = \frac{FV}{(1+r)^n} = \frac{2,000,000}{(1+10\%)^2} = 1,652,893$$

Example

- In November 1950, the Standard and Poor's 500 index was at 19.51. In November 2009, it was at 1,095.63. What was the annual rate of return over the 59 years ignoring dividend payouts?

$$r = [(\frac{FV}{PV})^{\frac{1}{n}} - 1] = (\frac{1095.63}{19.51})^{\frac{1}{2009-1950}} - 1 = 7.06\%$$

- You want to go to Thailand to live for two years and study international economics and business in a local school. You estimate that you will need 20,000 dollars for the two years. Currently, you only have 10,000 dollars, and the annual interest rate is 7%. How long should you wait until you leave?

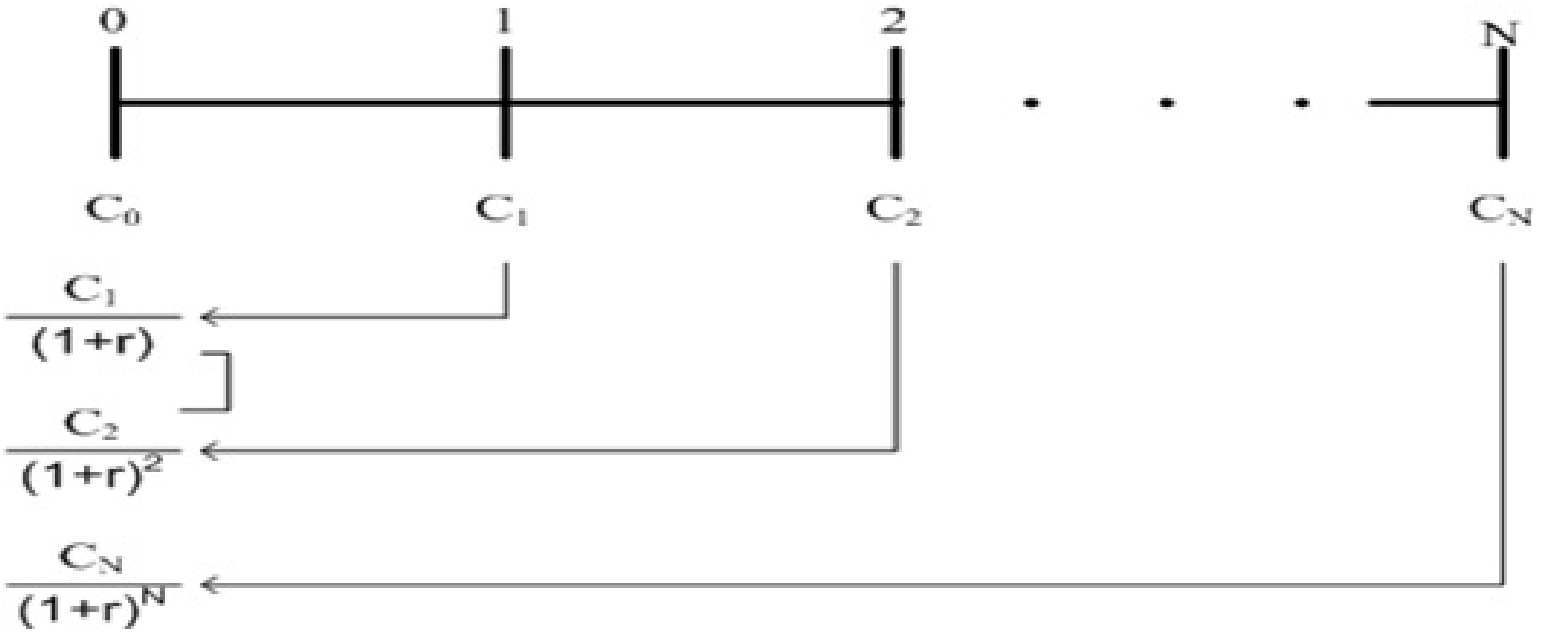
$$n = [\frac{\ln(\frac{FV}{PV})}{\ln(1+r)}] = [\frac{\ln(\frac{20{,}000}{10{,}000})}{\ln(1+0.07)}] = 10.24$$

Present Value of Multiple Cash Flows

- How do we determine the present value of multiple future cash flows?
  - Example: How much would you pay to receive two payments, \$10,000 the next year and \$20,000 in 3 years?
- We have to work out the Present Value of each cashflow separately, given the number of time periods between the cashflow and the present date.
- Think of each payment as a separate certified deposit. How much would you have to pay for this pair of certified deposits? We can simply add the separate Present Values of each future cashflow together, to derive the Present Value of the combination of them. This is called "value additivity".
  - $PV(CD_1 + CD_2) = PV(CD_1) + PV(CD_2)$.
  - $PV = \frac{10,000}{1+0.06} + \frac{20,000}{(1+0.06)^3}$

Present Value of Multiple Cash Flows

- In general:



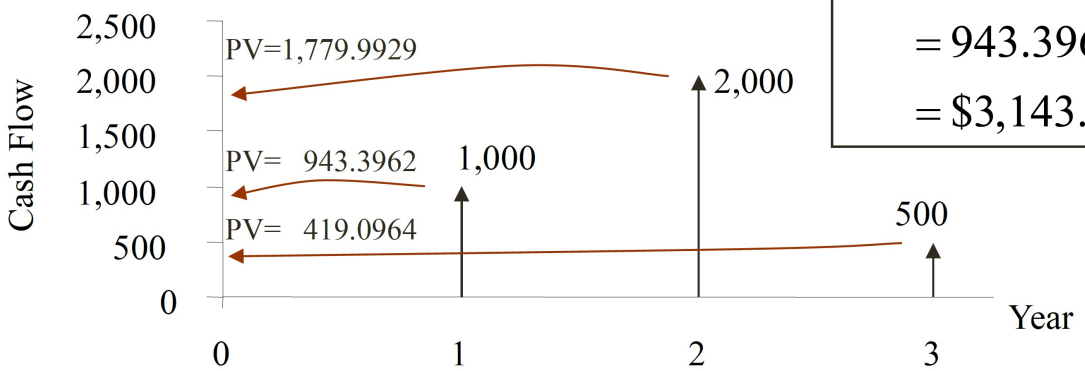
$$PV = CF_0 + \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \cdots + \frac{CF_N}{(1+r)^N}$$

Multiple Cash Flow Example

- Receive $1,000 next year (i.e., year 1)
- Receive $2,000 in 2 years
- Receive $500 in 3 years
- Find PV at 6% interest

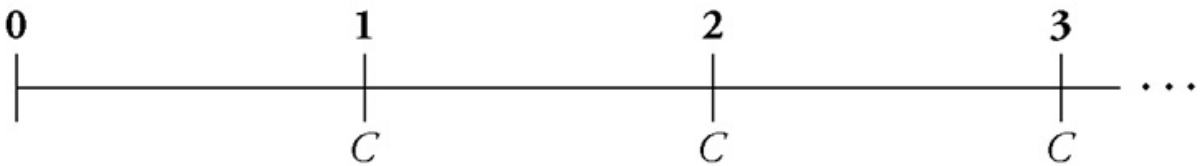
Year	CF
0	0
1	1,000
2	2,000
3	500

$$PV = \frac{1,000}{1.06} + \frac{2,000}{1.06^2} + \frac{500}{1.06^3}$$
$$= 943.3962 + 1,779.9929 + 419.0964$$
$$= \$3,143.20$$



Perpetuity

- Definition: A Perpetuity is a stream of fixed cash flows with an infinite life.
- Here is the timeline for a constant perpetuity:



- Examples:
  - Award/Scholarship funds
  - Consol bond: First issued by United Kingdom in 1752. The principal is never repaid. You lend a certain amount today, and the borrower gives you a fixed cash flow in perpetuity.

Perpetuity

- The present value of a perpetuity:

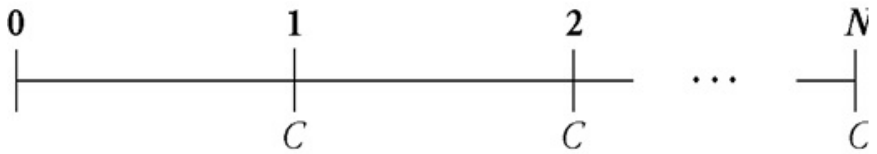
$$
\begin{aligned}
PV &= \frac{c}{(1+r)} + \frac{c}{(1+r)^2} + \frac{c}{(1+r)^3} + \cdots \\
&= c \cdot [\frac{1}{(1+r)} + \frac{1}{(1+r)} \cdot (\frac{1}{1+r}) + \frac{1}{(1+r)} \cdot (\frac{1}{1+r})^2 + \cdots] \\
&= c \cdot \frac{\frac{1}{(1+r)}}{1 - \frac{1}{(1+r)}} \\
&= \frac{c}{r}
\end{aligned}
$$

Perpetuity: Example

- IronClad produces steel bolts for the military. The contract it negotiated with the military states that it has the right to sell bolts to the military in perpetuity. It has decided to raise capital by issuing preferred stock that pays a 5 dollar dividend every quarter. The quarterly discount rate is 3%. Eva is trading preferred stock shares. She notices that IronClad's preferred stock is trading at \$150. Should she buy the stock?
- PV of Preferred Stock $= \dfrac{5}{0.03} = \$166.67$
- $\$166.7 > \$150$ Buy!

Valuing Finite Level Cash Flows: Annuities

- An **annuity** is a series of constant or level cash flows that occur at the end of each period for a **specified** number of periods.
  - All cash flows (C) are equal
  - None is skipped
  - N periods



- Examples
  - Fixed Rate Mortgages
  - Auto Loans
  - These are effectively annuities paid, rather than received, by the consumer.

Valuation of Annuity

- Annuity Present Value (APV):

$$APV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + ... + \frac{C}{(1+r)^n}$$
$$= \frac{C}{1+r} \cdot \frac{(\frac{1}{1+r})^n - 1}{\frac{1}{1+r} - 1}$$
$$= \frac{C}{r}[1 - (1+r)^{-n}]$$

- Annuity Future Value (AFV)$= APV \cdot (1+r)^n = \frac{C}{r}[(1+r)^n - 1]$
- Cosmo Kramer's "make your own pie" business is in a boom. The consulting firm Kramaerica LLP estimates the cash flow from the business will be $250,000$ annually and can last 10 years. Applying a discount rate of 8%, what is the present value of the "make your own pie" business?
  - $APV = \frac{250,000}{0.08}[1 - \frac{1}{(1+0.08)^{10}}] = \$1,677,520$

Mortgage Repayments

- Mortgages are a very familiar form of annuity. Suppose you have a 30-year mortgage (360 months) and a monthly rate of 1%. What is your monthly payment? Suppose you borrow $100,000.
- Key: the PV of the annuity at the start of the mortgage is equal to the amount loaned by the bank.

$$APV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + ... + \frac{C}{(1+r)^n}$$

$$100,000 = \frac{C}{(1+0.01)} + \frac{C}{(1+0.01)^2} + ... + \frac{C}{(1+0.01)^{360}}$$

$$100,000 = \frac{C}{0.01}[1-(1+0.01)^{-360}]$$

$$C = \$1,028.59$$

Definition of Internal Rate of Return

- Consider a project which lasts for n periods. Suppose the project generates cash flows c_i $(i = 0, 1, 2, \cdots, n)$ at time i. Suppose further that the interest rate is r. We can define the net present value of the project:

$$NPV = c_0 + \frac{c_1}{(1+r)} + \frac{c_2}{(1+r)^2} + \cdots + \frac{c_n}{(1+r)^n}$$

- As an investor, we should always invest on a project with positive net present value.
- A different criterion for investors is based on the internal rate of return(IRR), defined as the discount rate that makes the present value of all payments of a project to 0:

$$0 = c_0 + \frac{c_1}{(1+IRR)} + \frac{c_2}{(1+IRR)^2} + \cdots + \frac{c_n}{(1+IRR)^n}$$

Internal Rate of Return Rule

- The internal rate of return rule:
  - Accept project if $IRR >$ required rate of return (or opportunity cost of capital)
  - The benchmark return (required rate of return) is the discount rate that would be used to compute an investment's NPV.
- An investment project has an initial outlay of \$50,000, and returns in the end of year 1 and 2 are \$30,000 and \$40,000 respectively. Find the associated internal rate of return.

$$-50000 + \frac{30000}{(1+r)} + \frac{40000}{(1+r)^2} = 0$$

$$s = \frac{1}{1+r} \Rightarrow 40000s^2 + 30000s - 50000 = 0$$

$$4s^2 + 3s - 5 = 0 \Rightarrow s \approx 0.804$$

$$\Rightarrow r = \frac{1}{s} - 1 \approx 0.243 = 24.3\%$$

- If your required rate of return is $R = 15\%$, you'd accept the project.
- But for $R = 30\%$, you'd reject the project.